Restricted reciprocal-permutation sums and hollow Cauchy matrices

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Abstract

We prove the two restricted reciprocal-permutation congruences for every odd prime and every parameter prime to it, and the quadratic-residue derangement congruence. For parameters divisible by the prime, the signed formula holds for all primes at least five; at three its exact correction is $-a$ modulo nine. Thus the literal conjecture has exactly the two exceptional classes $p = 3$, $a \equiv 3, 6 \pmod 9$. The proof uses a first-order deformation of a hollow Cauchy matrix.

Source and scope. The target is Sun's Conjecture 4.4 in [1], page 26, reproduced in MathDB 351647. All congruences between rationals are in $\mathbb{Z}_{(p)}$, the rationals whose reduced denominators are prime to p. The source allows every integer a in the signed formula. We retain that quantifier and identify all exceptions; the result below also proves the remaining, unrestricted prime ranges.

Theorem 1. *Let p be an odd prime, $n = p - 1$, and $a \in \mathbb{Z}$. Define*

$$M_p(a)_{ij} = \begin{cases} (a+ij)^{-1}, & p \nmid a+ij, \\ 0, & p \mid a+ij, \end{cases} \qquad 1 \le i, j \le n.$$

If $p \nmid a$, then

$$\det M_p(a) \equiv \left(\frac{a}{p}\right) \frac{3 - a^{p-1}}{2} \pmod{p^2}, \tag{1}$$

$$\operatorname{per} M_p(a) \equiv (-1)^{(p+1)/2} \frac{3 - a^{p-1}}{2} \pmod{p^2}. \tag{2}$$

If $p \mid a$, the signed congruence (1) holds when $p \ge 5$. For $p = 3$ and $3 \mid a$, instead

$$\det M_3(a) \equiv -a \pmod 9. \tag{3}$$

Finally, if $p \equiv 1 \pmod 4$ and $m = (p-1)/2$, then

$$\sum_{\tau \in D(m)} \prod_{j=1}^{m} \frac{1}{j^2 - \tau(j)^2} \equiv \big((m/2)!\big)^{-2} \pmod p. \tag{4}$$

Here $D(m)$ is the set of permutations without fixed points.

The determinant and permanent of $M_p(a)$ are exactly the signed and unsigned restricted sums: setting a forbidden entry to zero deletes precisely the forbidden permutations.

Prior ingredients. Cycle cancellation for reciprocal differences already appears in Guo–Li–Tao–Wei [2, Lemma 3.1]. The resulting permanent–determinant identity, and the canonical determinant modulo p^2, appear in Yang–Zhang [3, Proposition 12 and Lemma 15]. We give self-contained proofs and claim no separate problem credit for these ingredients. The quadratic-residue calculation is also closely related to the root-of-unity evaluations in [2].

Lemma 2. *For an even integer N and pairwise distinct nodes $x_1, \dots, x_N$, let $C(x)$ have off-diagonal entries $(x_i - x_j)^{-1}$ and diagonal zero. As a rational identity,*

$$\operatorname{per} C(x) = (-1)^{N/2} \det C(x).$$

Proof. Fix a set of $\ell \geq 3$ nodes and distinguish one of them, z. Every oriented cycle on the set is obtained uniquely by inserting z in one edge (u, v) of a cycle on the other $\ell - 1$ nodes. The factor by which the cycle product changes is

$$\frac{u - v}{(u - z)(z - v)} = \frac{1}{u - z} - \frac{1}{v - z}.$$

Summing this over the edges of any fixed contracted cycle gives zero. Thus the sum of products over all cycles on a given set of at least three nodes is zero. In either permutation expansion, group permutations by the sets supporting their cycles. Fixed points contribute zero; a block of size at least three cancels. The remaining permutations are perfect matchings, all with sign $(-1)^{N/2}$. This proves the identity. It remains valid over any field whenever the node differences are units. □

Lemma 3. *For $C_p = C(1, 2, \dots, p-1)$, one has $\det C_p \equiv 1 \pmod{p^2}$.*

Proof. Write $n = p - 1$, $P(X) = \prod_{j=1}^{n}(X - j)$, $w_j = 1/P'(j)$, $W = \operatorname{diag}(w_j)$, and $X = \operatorname{diag}(1, \dots, n)$. The matrix D that sends the values of a polynomial of degree below n to the values of its derivative has, by Lagrange interpolation,

$$D_{ij} = \frac{w_j}{w_i(i - j)} \quad (i \neq j), \qquad D_{ii} = \sum_{j \neq i} \frac{1}{i - j} = H_{i-1} - H_{n-i},$$

where $H_0 = 0$ and $H_r = \sum_{j=1}^{r} 1/j$. Set $L = D + X^{-1}$. The matrix $XL = XD + I$ represents $f \mapsto xf' + f$, whose eigenvalues on $1, x, \dots, x^{n-1}$ are $1, \dots, n$. Therefore

$$\det L = \frac{n!}{\det X} = 1.$$

Conjugation by W shows that L is similar to $C_p + \operatorname{diag}(u_i)$, where $u_i = H_i - H_{p-1-i}$. Since $H_{p-1} \equiv 0 \pmod{p}$ and $\sum_{j=p-i}^{p-1} 1/j \equiv -H_i \pmod{p}$, every u_i is in $p\mathbb{Z}_{(p)}$. In expanding $\det(C_p + \operatorname{diag}(u_i))$, each term linear in a u_i has an odd-order skew-symmetric principal minor as coefficient, so that coefficient is zero. Terms containing at least two u_i lie in $p^2\mathbb{Z}_{(p)}$. Consequently $\det C_p \equiv \det L = 1 \pmod{p^2}$. □

Lemma 4. *Let $x_i = i$ for $1 \leq i \leq p - 1$, and let $y_j = j + \delta_j$, where $\delta_j \in p\mathbb{Z}_{(p)}$. Define $H(x, y)$ to have zero diagonal and off-diagonal entries $(x_i - y_j)^{-1}$. Then, with $\chi = (-1)^{(p-1)/2}$,*

$$\det H(x, y) \equiv 1 - \frac{1}{2} \sum_{j=1}^{p-1} \frac{\delta_j}{j} \pmod{p^2}, \tag{5}$$

$$\operatorname{per} H(x, y) \equiv \chi \det H(x, y) \pmod{p^2}. \tag{6}$$

Proof. Put $F(x,y) = \det H(x,y)$ and $G(x) = F(x,x) = \det C(x)$. Since $H(y,x) = -H(x,y)^T$ and $n = p-1$ is even, $F(x,y) = F(y,x)$. On the diagonal this gives

$$F_{y_j}(x,x) = F_{x_j}(x,x) = \tfrac{1}{2} G_{x_j}(x).$$

We compute the last derivative at $r = (1, \dots, p-1)$ over $\mathbb{F}_p$. The rational function G is symmetric and homogeneous of degree $-n$. For $g \in \mathbb{F}_p^\times$, multiplication by g permutes the coordinates of r. If $\pi_g(j)$ is the index whose residue is gj, symmetry and homogeneity yield

$$G_{x_{\pi_g(j)}}(r) = g^{-n-1} G_{x_j}(r), \qquad r_{\pi_g(j)} G_{x_{\pi_g(j)}}(r) = g^{-n} r_j G_{x_j}(r) = r_j G_{x_j}(r).$$

Thus all $r_j G_{x_j}(r)$ have the same value. Euler's identity says their sum is $-nG(r)$. Since n is nonzero in $\mathbb{F}_p$, their common value is $-G(r) = -1$, by Lemma 3. It follows that $F_{y_j}(r,r) = -1/(2j)$ in $\mathbb{F}_p$. The constant term is $F(r,r) \equiv 1 \pmod{p^2}$. Taylor expansion in the δ_j now proves (5).

For completeness set $R(x,y) = \operatorname{per} H(x,y) - \chi \det H(x,y)$. It too is symmetric in x, y, and $R(x,x) = 0$ identically by Lemma 2. Differentiation gives $R_{x_j}(x,x) + R_{y_j}(x,x) = 0$, while symmetry makes the two terms equal. They therefore both vanish. Its expansion at (r,r) has neither constant nor linear terms, proving (6). All denominators in these expansions are products of off-diagonal node differences, which are units modulo p; all terms of degree at least two are consequently in $p^2 \mathbb{Z}_{(p)}$. □

Proof of the unit-parameter formulas. Assume $p \nmid a$. Let σ be the permutation of $1, \dots, n$ satisfying $\sigma(j) \equiv -a/j \pmod{p}$, and put

$$y_j = -\frac{a}{\sigma(j)}, \qquad \delta_j = y_j - j, \qquad B = (p-1)!, \qquad A = a^{p-1}.$$

Then $\delta_j \in p\mathbb{Z}_{(p)}$. For the column-permuted matrix,

$$M_p(a)_{i,\sigma(j)} = \frac{1}{\sigma(j)} H(x,y)_{ij}.$$

This equality also holds on the diagonal: both entries are forced to zero there. It follows that

$$\det M_p(a) = \frac{\operatorname{sgn}\sigma}{B} \det H(x,y), \qquad \operatorname{per} M_p(a) = \frac{1}{B} \operatorname{per} H(x,y). \tag{7}$$

The exact product identity

$$\prod_{j=1}^{n} \frac{y_j}{j} = \frac{A}{B^2}$$

implies $\sum_j \delta_j / j \equiv A/B^2 - 1 \pmod{p^2}$, so

$$\det H(x,y) \equiv \frac{3 - A/B^2}{2} \pmod{p^2}.$$

Write $B = -1 + pw$ and $A = 1 + pq$, using Wilson's and Fermat's theorems. Then

$$B^{-1} \frac{3 - A/B^2}{2} \equiv (-1 - pw)\left(1 - pw - \frac{pq}{2}\right) \equiv -1 + \frac{pq}{2} = -\frac{3 - A}{2} \pmod{p^2}.$$

To evaluate $\operatorname{sgn}\sigma$, inversion on $\mathbb{F}_p^\times$ fixes $1, -1$ and has $(p-3)/2$ transpositions, hence sign $-\chi$. Multiplication by $b \in \mathbb{F}_p^\times$ has sign (b/p): the ratio of Vandermonde products gives its sign modulo p as $b^{n(n-1)/2} = (b^{(p-1)/2})^{p-2} = (b/p)$, and the signs $1, -1$ are distinct modulo an odd prime. Taking $b = -a$ gives $\operatorname{sgn}\sigma = (-\chi)(-a/p) = -(a/p)$. Substitution in (7), together with Lemma 4, proves (1) and (2).

Parameters divisible by the prime. If $p \mid a$, every entry is allowed and $M_p(a)_{ij} \equiv 1/(ij) \pmod p$. For each $i > 1$, subtract i^{-1} times the first row from row i. All entries in those $n-1$ rows now lie in $p\mathbb{Z}_{(p)}$, so $\det M_p(a) \in p^{n-1}\mathbb{Z}_{(p)}$. For $p \geq 5$, this is zero modulo p^2, as is the right side of (1). For $p = 3$ and $3 \mid a$, direct calculation gives

$$\det M_3(a) = \frac{1}{(a+1)(a+4)} - \frac{1}{(a+2)^2} = \frac{-a}{(a+1)(a+4)(a+2)^2} \equiv -a \pmod 9.$$

The denominator is 1 modulo 3, and the numerator is divisible by 3, which justifies the last reduction modulo 9. Thus the original zero prediction fails exactly for $a \equiv 3, 6 \pmod 9$, and succeeds for $a \equiv 0 \pmod 9$.

Proof of the quadratic-residue formula. Work over $\mathbb{F}_p$. Since $p \equiv 1 \pmod 4$, the integer $m = (p-1)/2$ is even. The nodes $x_j = j^2$, $1 \leq j \leq m$, are distinct and are precisely the roots of $P(T) = T^m - 1$. Let $Q = C(x)$ and $X = \operatorname{diag}(x_j)$. The same Lagrange differentiation matrix as in Lemma 3 has diagonal

$$D_{jj} = \frac{P''(x_j)}{2P'(x_j)} = \frac{m-1}{2x_j}.$$

Writing $c = (m-1)/2$ in $\mathbb{F}_p$, diagonal conjugation therefore makes Q similar to $D - cX^{-1}$. On polynomials of degree below m, XD represents $f \mapsto xf'$, with eigenvalues $0, 1, \ldots, m-1$. Consequently

$$\det Q = \frac{\prod_{d=0}^{m-1}(d-c)}{\prod_{j=1}^{m} x_j} = \frac{(-1)^{m/2}((m-1)!!)^2}{2^m (m!)^2}.$$

By Lemma 2 and $m! = 2^{m/2}(m/2)!(m-1)!!$, this gives

$$\operatorname{per} Q = \frac{1}{2^{2m}((m/2)!)^2} = \frac{1}{((m/2)!)^2} \quad \text{in } \mathbb{F}_p,$$

where $2^{2m} = 2^{p-1} = 1$. The zero diagonal makes $\operatorname{per} Q$ the sum in (4), completing Theorem 1. □

Verification and limits. The accompanying standard-library Python verifier evaluates the original matrices using fraction-free integer determinants and subset dynamic programming for permanents. It checks every parameter modulo p^2 for $p = 3, 5, 7, 11, 13$, additional larger-prime determinants, the quadratic-residue formula, and arbitrary first-order column deformations. Exact rational permutation enumeration supplies a separate small-matrix cross-check. These are corroborating computations; the preceding proof establishes the infinite ranges. The package has author-agent review, not external peer review.

Task 2. A lift modulo p^3. Determine the second-order deformation of $H(x, y)$ and the first possible correction to (1)–(2) modulo p^3. Express the answer using node displacements and harmonic sums.

Task 3. Changing integer representatives. Replace $1, \ldots, p-1$ by arbitrary integer representatives of $\mathbb{F}_p^\times$ in the row and column sets. Determine exactly which linear displacement statistics govern the resulting sums modulo p^2.

Task 4. Other multiplicative subgroups. For each even divisor m of $p-1$, and a coset of the subgroup of order m, evaluate the hollow reciprocal-difference permanent. Determine which such evaluations lift from $\mathbb{F}_p$ to $\mathbb{Z}/p^2\mathbb{Z}$.

Task 5. More than one forbidden residue. Replace the denominator $a+ij$ by a polynomial with two or more forbidden residue classes. Identify conditions under which a permanent–determinant relation survives to first order.

References

[1] Z.-W. Sun, *Arithmetic properties of some permanents*, arXiv:2108.07723v7 (2022), Conjecture 4.4, p. 26.

[2] X. Guo, X. Li, Z. Tao and T. Wei, *The eigenvectors-eigenvalues identity and Sun's conjectures on determinants and permanents*, arXiv:2206.02592 (2022); *Linear Multilinear Algebra* **72** (2024), 1071–1077, doi:10.1080/03081087.2023.2172380. The lemma locator above refers to the arXiv version.

[3] Y. Yang and Y. Zhang, *Sun-type determinant and permanent congruences*, arXiv:2605.19502v3 (2026).